



Alimentary system module

Liver, pancreas and biliary system

SGL 3

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Autoimmune cholangiopathies

- **It includes two disorders:**
 - **Primary biliary cholangitis**
 - **Sclerosing cholangitis.**

Primary biliary cholangitis (PBC)

- **Is an autoimmune disease whose principal feature is non-suppurative, inflammatory destruction of small- and medium-sized intrahepatic/interlobular bile ducts.**
- **Large intrahepatic ducts and the extrahepatic biliary tree are not involved.**
- **The retention of bile salts due to bile duct injury leads to secondary hepatocellular injury that can eventually progress to cirrhosis.**
- **PBC is primarily a disease of middle-age women.**
- **Its peak incidence is between 40 and 50 years of age.**
- **Both environmental and genetic factors are important in its pathogenesis.**

Primary biliary cholangitis (PBC)

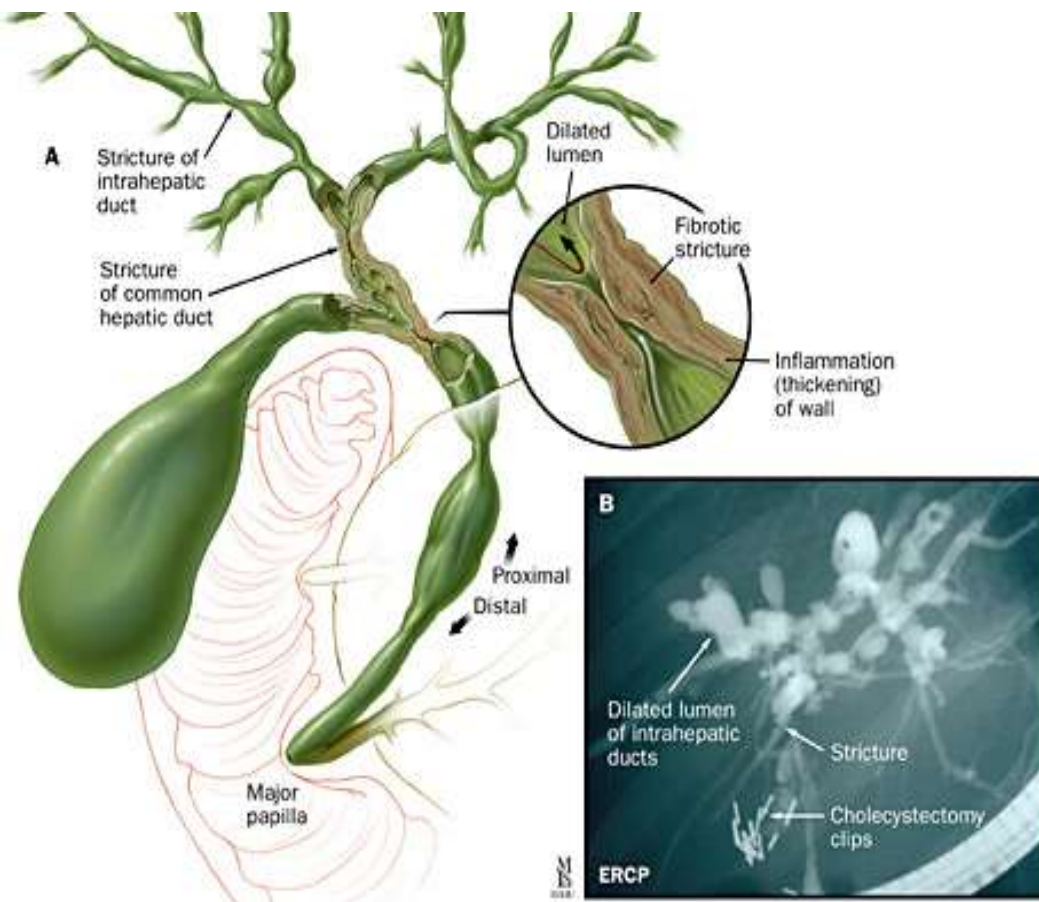
- **Antimitochondrial antibodies** are present in 90% to 95% of patients.
- In the absence of treatment, the disease follows one of two paths to end stage disease:
 - In the first, more common pathway, slowly leading to cirrhosis and eventually to profound cholestasis.
 - Alternatively, some patients develop prominent portal hypertension rather than severe cholestasis.

Fortunately, both of these outcomes are now rarely seen because of effective therapy.

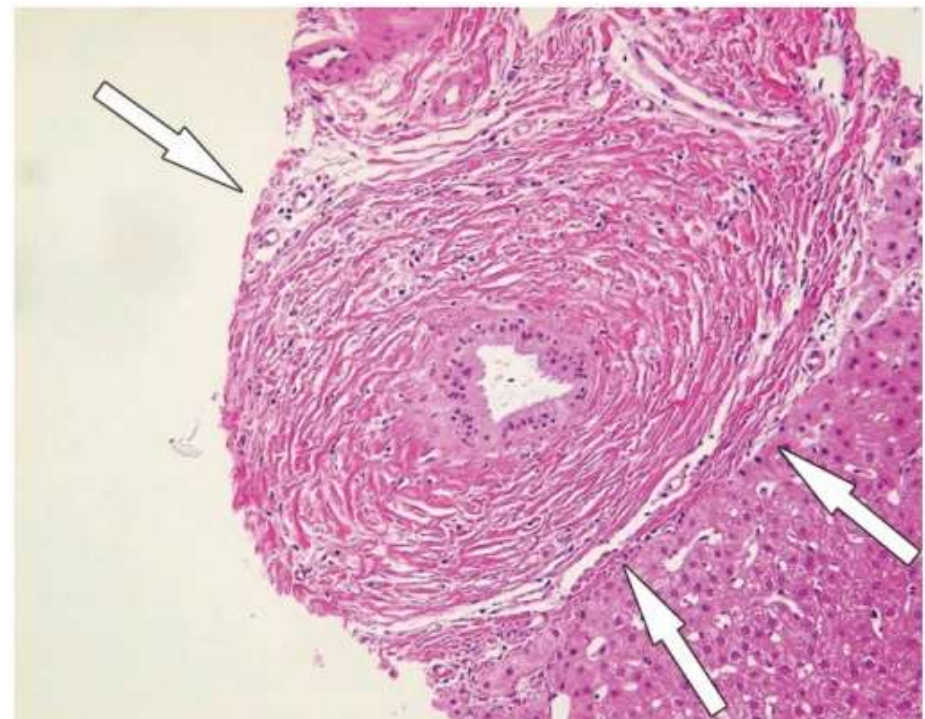
Primary sclerosing cholangitis (PSC)

- **It is characterized by inflammation and obliterative fibrosis of intrahepatic and extrahepatic bile ducts, leading to dilation of preserved segments.**
- **Irregular biliary strictures and dilations cause the characteristic “beading” of the intrahepatic and extrahepatic biliary tree seen by MRI.**
- **Inflammatory bowel disease, most commonly ulcerative colitis, coexists in approximately 70% of individuals with PSC.**
- **PSC tends to occur in the third through fifth decades of life.**
- **It has a 2 : 1 male predominance.**

Primary sclerosing cholangitis (PSC)



Classic histologic finding is "onion skin" fibrosis around affected bile ducts



Primary sclerosing cholangitis (PSC)

- Several autoantibodies are present in PSC:
 - **Anti-smooth muscle antibodies**, in approximately 75% of patients
 - Antinuclear antibodies.
 - Antibodies directed against cytoplasmic and nuclear antigens of neutrophils (ANCA) are found.
- Progression to cholangiocarcinoma may occur.

Table 14.5 Main Features of Primary Biliary Cholangitis and Primary Sclerosing Cholangitis

Parameter	Primary Biliary Cholangitis	Primary Sclerosing Cholangitis
Age	Median age 50 years	Median age 30 years
Sex	90% female	70% male
Clinical course	Progressive	Unpredictable, but progressive—may progress to cholangiocarcinoma
Associated conditions	Sjögren syndrome (70%)	Inflammatory bowel disease (70%)
	Scleroderma (5%)	Autoimmune pancreatitis
	Thyroid disease (20%)	IgG4 related fibrosing diseases
Serology	95% AMA-positive	0%–5% AMA-positive (low titer)
	20% ANA-positive	6% ANA-positive
	40% ANCA-positive	65% ANCA-positive
Radiology	Normal	Strictures and beading of large bile ducts; pruning of smaller ducts
Duct lesion	Florid duct lesions and loss of small ducts only	Inflammatory destruction of extrahepatic and large intrahepatic ducts; fibrotic obliteration of medium and small intrahepatic ducts

AMA, Antimitochondrial antibody; ANA, antinuclear antibody; ANCA, antineutrophil cytoplasmic antibody.

Gallstones

There are of two main types:

- **Cholesterol stones, containing crystalline cholesterol monohydrate.**
- **Pigment stones, made of bilirubin calcium salts.**

Gallstones/ Risk factors

Cholesterol Stones

Advancing age
Female sex hormones
 Female sex
 Oral contraceptives
 Pregnancy
Obesity and insulin resistance
Rapid weight reduction
Gallbladder stasis
Inborn disorders of bile acid metabolism
Dyslipidemia syndromes

Pigment Stones

Chronic hemolysis (e.g., sickle cell anemia, hereditary spherocytosis)
Biliary infection
Gastrointestinal disorders: ileal disease (e.g., Crohn disease), ileal resection or bypass, cystic fibrosis with pancreatic insufficiency

Cholesterol gallstone

Pathogenesis:

- Cholesterol is rendered water soluble by aggregation with bile salts and lecithins.
- When cholesterol concentrations exceed the solubilizing capacity of bile (supersaturation), cholesterol can no longer remain dispersed and crystallizes out of solution.
- Cholesterol gallstone formation is enhanced by:
 - Hypomotility/ stasis of the gallbladder, which promotes crystal nucleation.
 - Mucus hypersecretion, which traps crystals, thereby enhancing their aggregation into stones.

Cholesterol gallstone

Cholesterol GS could be either:

- **Pure:** less common, about 10% of cholesterol GS
 - Composed of at least 90% cholesterol.
 - Pale yellow, round to ovoid.
 - Most often are multiple, rarely single.
 - Often radiolucent.
- **Mixed:** More common
 - Composed of at least 60% cholesterol, the rest is usually calcium salts.
 - Radiopaque because of their calcium content.



Pigment gallstones

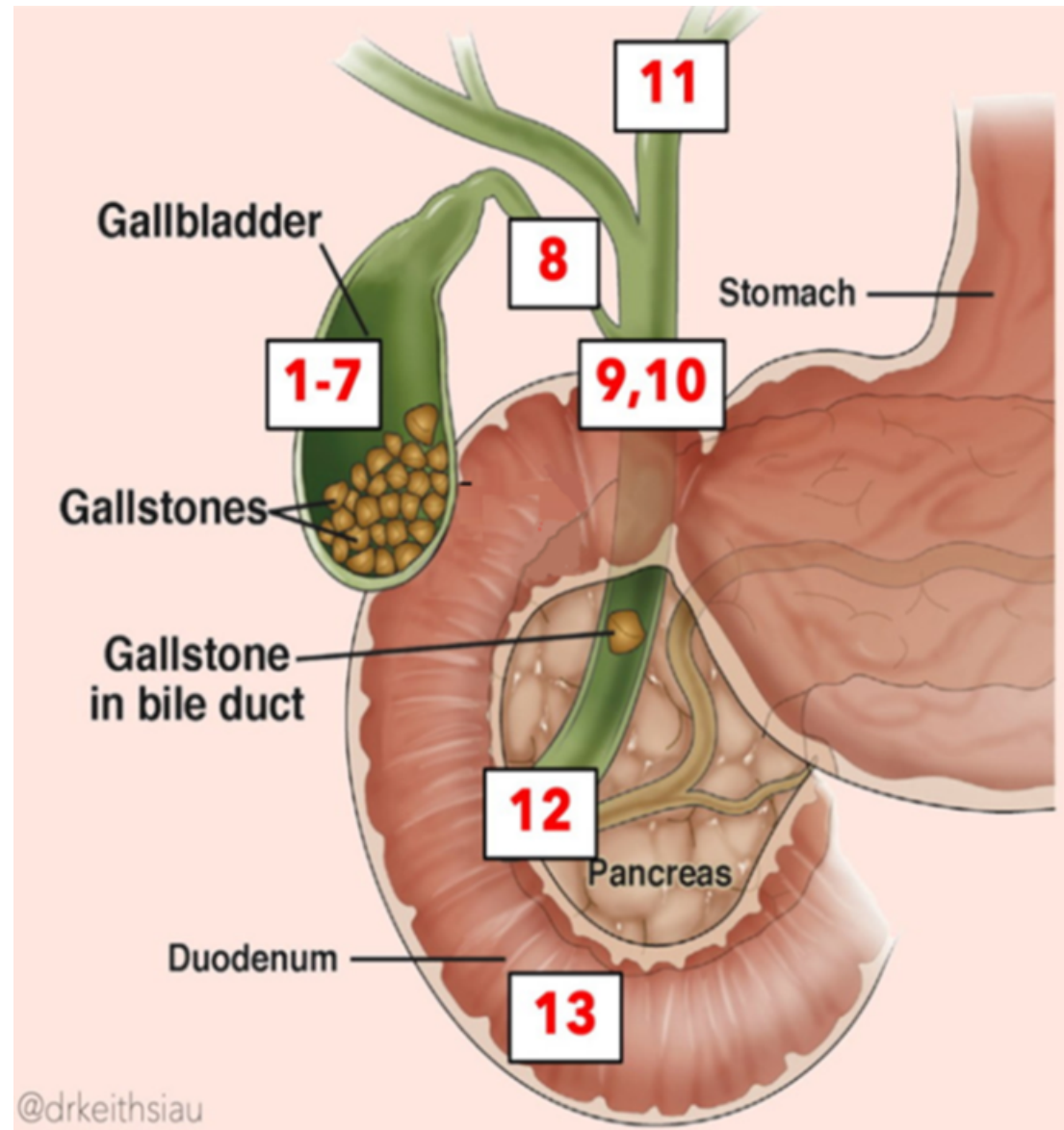
- Are made up of insoluble calcium bilirubinate salts.
- Pathogenesis: they form when bile contains a high concentration of unconjugated bilirubin, as may occur in patients with chronic hemolysis or with certain infections of the biliary tract.
- Pigmented stones may be either;
 - Black color:
 - Usually formed in sterile gallbladder bile, associated with intravascular hemolysis.
 - Usually small, numerous, and fragile to the touch.
 - 50% to 75% of black stones are radiopaque
 - Brown color:
 - Found in infected intrahepatic or extrahepatic ducts.
 - Single or few in number and to have a soft, greasy, soap-like consistency.
 - Usually radiolucent.

Pigment gallstones



Outcomes and complications of GS

1. Biliary colic.
2. Cholecystitis.
3. Mucocele.
4. Empyema.
5. Fistula.
6. Perforation.
7. Gallbladder carcinoma.
8. Common hepatic duct obstruction.
9. Choledocholithiasis.
10. Cholangitis.
11. Intrahepatic stone.
12. Pancreatitis.
13. Gall stone ileus.



Cholecystitis

- Inflammation of the gallbladder may be:
 - Acute cholecystitis which in term could be either:
 - Acute calculus Cholecystitis.
 - Acute acalculus Cholecystitis.
 - Chronic Cholecystitis
 - Acute on chronic, acute exacerbation of the inflammatory process in a patient with known chronic cholecystitis.
- Almost always occurs in association with gallstones.
- Its epidemiologic distribution closely parallels that of gallstones.

Acute Cholecystitis

Acute calculous Cholecystitis:

- Is the commonest form, seen in 90% of cases.
- Caused by gall stone causing obstruction of the gallbladder neck or cystic duct.
- Clinically, it can present with right upper abdominal pain, biliary colic, nausea, vomiting, low-grade fever, and leukocytosis.
- Complications include gangrene of the gallbladder, perforation, peritonitis, fistula formation and gallstone ileus.

Acute Cholecystitis

Acute acalculous cholecystitis

- Is thought to result from gallbladder stasis and ischemia leading to a local inflammatory response.
- Most cases occur in seriously ill patients.
- Some of the most common predisposing insults are as follows:
 - Major surgery
 - Severe trauma
 - Severe burns
 - Sepsis
- Has higher mortality rate because of the associated conditions and greater complication risk likely gangrene and perforation.

Chronic Cholecystitis

- **May occurs after repeated bouts of acute cholecystitis, or it may develops without an antecedent history of acute attacks.**
- **It is almost always associated with gallstones, rarely could be acalculous.**
- **Supersaturation of bile appears to predispose to both chronic inflammation and, in most instances, stone formation.**
- **Microorganisms, usually E. Coli and enterococci, can be cultured from the bile in about one-third of cases.**

Acute Calculous Cholecystitis:

Showed mucosal erythema, edema, congestion to extensive ulceration and necrosis. In addition to areas of patchy greenish-yellow purulent exudate.



Carcinoma of the gallbladder

- **Is the most common malignancy of the extrahepatic biliary tract.**
- **Risk factors include:**
 - **Gallstones: present in 95% of cases of GB carcinoma possibly as a result of chronic inflammation.**
 - **Carcinogenic derivatives of bile acids.**
 - **Primary sclerosing cholangitis.**
- **Clinically:**
 - **usually diagnosed at time of surgery for unrelated condition.**
 - **Presenting symptoms are insidious and typically indistinguishable from those associated with cholelithiasis.**

Carcinoma of the gallbladder

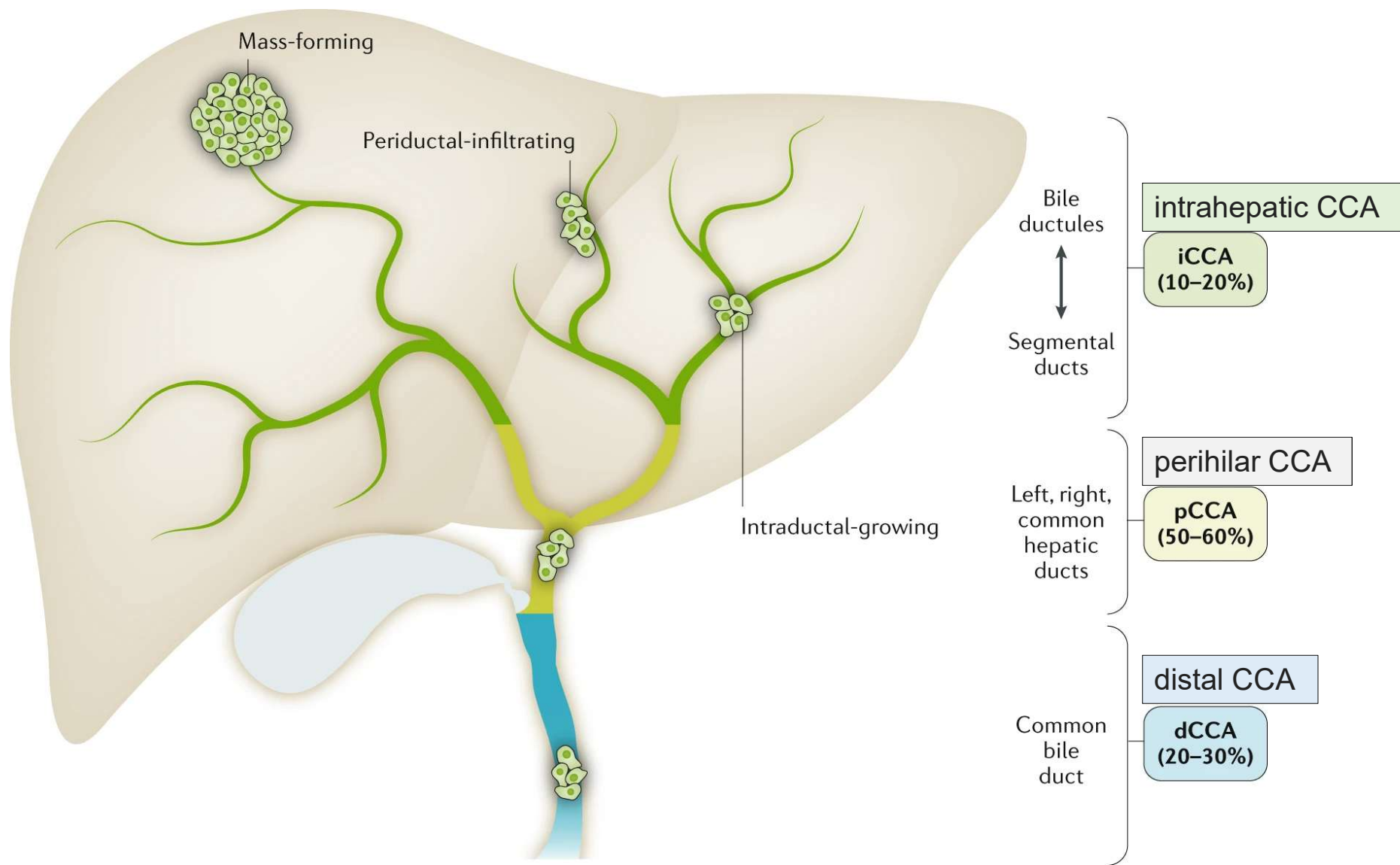
Pathology:

- Grossly GB carcinomas may show two patterns of growth:
 - Infiltrative: as a poorly defined area of diffuse wall thickening and induration.
 - Exophytic: grows as an irregular exophytic mass into the lumen.
- Microscopically:
 - Most GB carcinomas are adenocarcinomas.
 - About 5% are squamous cell carcinomas.



Cholangiocarcinoma CCA

- **Carcinoma arises from intrahepatic and extrahepatic bile ducts.**
- **Histologically, they are of adenocarcinoma type.**
- **The risk factors include conditions associated with chronic inflammatory process, like:**
 - **Infestation by liver flukes.**
 - **Primary sclerosing cholangitis**
 - **Intrahepatic gallstones**
 - **Hepatitis B and C and NAFLD.**
- **Classification of the type of CCA depends on its location.**



Intrahepatic CCA may present as a mass (white arrow) and/ or ductal infiltration pattern (black arrow)

